

ever. There is a vast digital divide that brings about the need to make these technologies as accessible as possible to bridge the gap between rural and urban areas in terms of employability and skill. Some of the technologies that play a huge role in contributing to the education and skill development sector are discussed next.

Cloud computing

Cloud computing brings teachers and students on the same platform and provides ease of sharing resources remotely with each other. Teachers can create online courses while students can give virtual exams in organisations like schools and colleges. Working professionals can also take these classes. Assignments are not just limited to documents but can also include uploading a video recording or taking pictures of work and sharing them with peers.

Virtual classrooms allow hiring remote trainers and also reduce infrastructural expenses. There are companies in the field of education like Tata Class Edge, Educomp, and Pearson, who are aiding teachers in classroom teaching.

The storage devices like solid-state drives (SSDs) and hard disc drives (HDDs) must be optimised for cloud computing applications. There is no requirement of purchasing and maintaining servers and data centres. The databases, storage, and other services can be accessed whenever required, thereby saving time.

Most cloud providers ensure a base level of security for their cloud infrastructure. Using a VPN can ensure data security.

The digital assistant Beacon, launched by UK-based Staffordshire University and hosted on the Microsoft Azure cloud computing platform, is capable of answering questions that students might have to help them in studying, checking their moods, and suggesting them activities. If a student is still struggling, the assistant sends alerts to staff members so that they can help.

AI and ML

Artificial intelligence (AI) and machine learning (ML) have transformed learning to improve its outcomes. Personalised learning caters



Immersive learning and training (Credit: <https://i.pinimg.com>)

to the individual needs of students by adopting different teaching methods to make them more comfortable for upcoming tests. This is done by gathering actionable insights for platforms by tracking various data points and keywords from previous feedback, engagement, and interaction levels.

This is necessary to conclude the strong and weak points of different methods. Even checking test papers, grading, paper setting, creating mark-sheets, and tracking progress can be done automatically and intelligently. Many institutes have started using onscreen evaluation systems to auto-calculate scores. Natural language processing (NLP) makes real-time language translation possible, enabling people who speak different languages to communicate with one another.

In Andhra Pradesh, ML was used to collect and analyse the data of students related to parameters such as academic performance, skills of teachers, and the like. The application could find patterns to predict outcomes like probable students who would drop out in 2018-19.

Robotics and 3D printing

As in-demand skills change from time to time, developing problem-solving skills is a must. What could be a better way to engage children than educational robots? Even if the purpose is not to teach coding or robotics, skills such as analytical thinking and teamwork can be developed during the creation, design, and operation of robots. Personal robots are also useful for children with autism to help them in learning, besides improving social interaction skills.

Robotics is also applied in streams

like engineering. Educational robotics kit LEGO Mindstorms has been integrated into curriculums at higher education institutions like MIT, University of Maryland, Tufts University, Trinity College, and the University of Manchester, among others. This is used mainly to assist in the mathematical analysis required to move a robot, such as using appropriate formulas and blocks for calculations.

Atal Tinkering Laboratories (ATL) established by the government of India in educational institutes across India have do it yourself (DIY) kits and equipment related to robotics like open source microcontroller boards and sensors, 3D printers, and more. As 3D printers are becoming more affordable, they are often found in classrooms (commonly Fused Deposition Modeling).

Physical manipulation of objects like historical artifacts, models of molecules and organs, and other 3D objects helps develop a better understanding and complements the usual theory-based curriculum. In the past, Macquarie University's Department of Educational Studies, together with the company Makers Empire, did research to conclude how 3D printing could boost creativity, critical thinking, and design skills among primary school students.

XR

Extended reality (XR)—a term that combines augmented (AR), mixed (MR), and virtual reality (VR)—offers a wide range of opportunities for immersive learning and training experiences. This is highly useful in industries like healthcare, manufacturing, and aerospace where accuracy and precision are crucial to prevent dangerous repercussions.

AR and VR can instruct employees when they actually need to use it in real time. Learners can interact with the environment and get a view of different angles for performing their tasks. This reduces the time taken to conceptualise an idea from a video or document and execute it.

Individuals learn by practising and